Lecture 14 (10/16)

Simpson's paradox and regression with many predictors

Last Class:

- Proved OVB and that ρ^2 never decreases as we add predictors
- Examples (violent crime and baseball)

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Today:

1) Assignment 2
2) General formula for $K \geq 2$ predictors.
3) Examples with more than two predictors

Assignment 2: Gaza casualty counts

Wyner argues the Gaza Health Ministry's daily casualty counts are fabricated; Pachter's blog post responds to his first claim.

1) In the tablet article, what are some of the statistical claims and are they justified? Is there evidence that "The total civilian casualty count is likely to be extremely overstated"?

Assignment 2: Gaza casualty counts

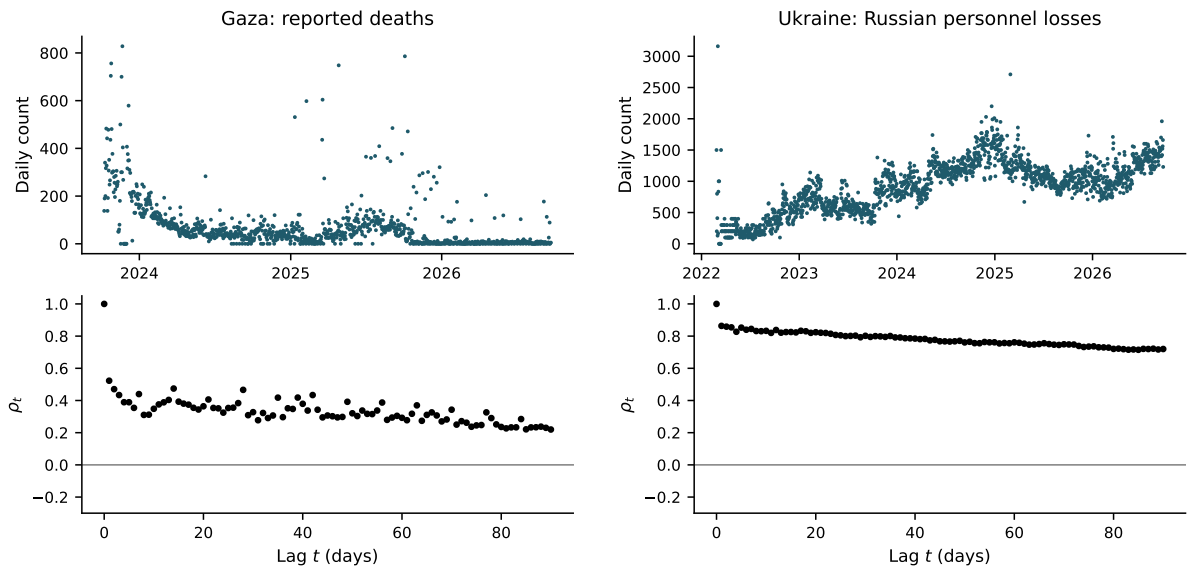
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2) Does the "Garden of forking paths" apply to this analysis?
3) Is the following plot of the lagged correlations of daily deaths in the Ukraine and Gaza dataset to be expected? Why or why not?



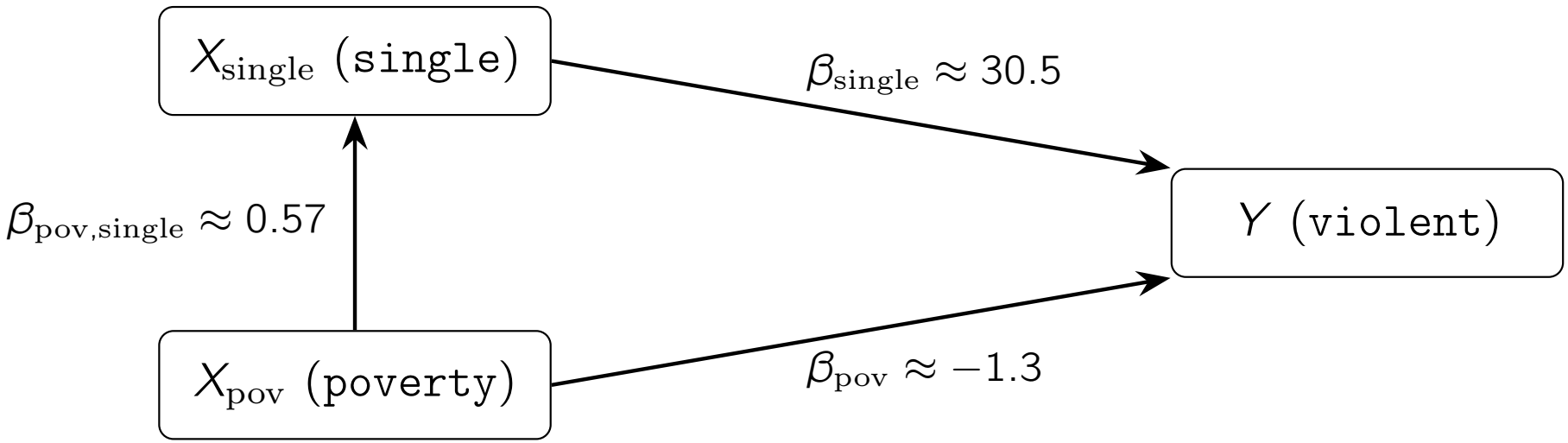
Crime data review

Recall from last class:

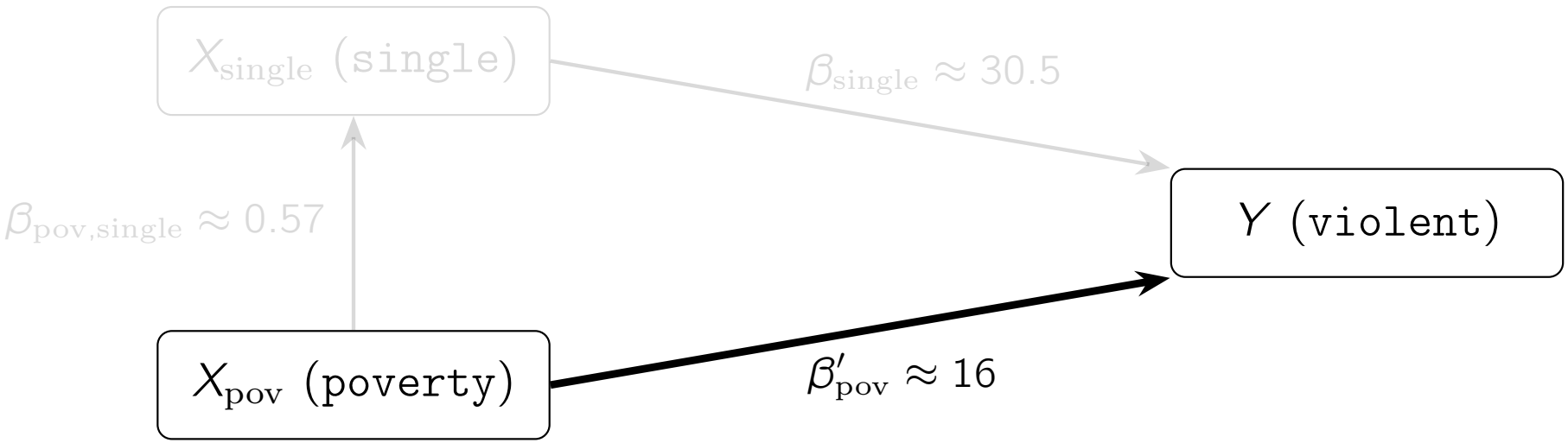
- Y (`violent`): violent crimes per 100,000 people;
- X_{pov} (`poverty`): % of people below the poverty line;
- X_{single} (`single`): % of family households headed by a single parent.

```
Dep. Variable:                violent   R-squared:                       0.093
                 coef    std err          t      P>|t|      [0.025      0.975]
const        172.9711    101.400      1.706      0.095     -30.906     376.849
poverty       15.9665      7.193      2.220      0.031       1.503      30.430
Dep. Variable:                violent   R-squared:                       0.436
                 coef    std err          t      P>|t|      [0.025      0.975]
const       -348.9463    122.900     -2.839      0.007    -596.054    -101.838
single        29.9911      4.920      6.095      0.000      20.098      39.884
Dep. Variable:                violent   R-squared:                       0.437
                 coef    std err          t      P>|t|      [0.025      0.975]
const       -344.7585    125.969     -2.737      0.009    -598.175     -91.342
poverty       -1.2906      6.573     -0.196      0.845     -14.514      11.933
single        30.5401      5.703      5.355      0.000      19.067      42.013
```

Leaving out single-parent households



Leaving out single-parent households



$$\underbrace{16}_{\beta'_{\text{pov}}} \approx \underbrace{-1.3}_{\beta_{\text{pov}}} + \underbrace{0.57 \times 30.5}_{\beta_{\text{pov,single}}\beta_{\text{single}} \approx 17.4}$$

`poverty` picks up the association that runs through single-parent households.

Who is the better hitter?

Batting average = hits / at bats, for two New York Yankees:

	1995	1996	both years
Derek Jeter	$12/48 = .250$	$183/582 = .314$	
David Justice	$104/411 = .253$	$45/140 = .321$	

Who is the better hitter?

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Justice has the higher average *in each year*, but Jeter has the higher average *overall*. This is an example of Simpson's paradox.

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Question

Explain this in terms of omitted variable bias. What is the omitted variable?

Conditional means and the omitted year

Let $Y = 1$ for a hit and $Y = 0$ otherwise, $X_{\text{jeter}} = 1$ for Jeter (0 for Justice), and $X_{\text{year}} = 1$ for 1996 (0 for 1995). We consider a model

$$E[Y \mid X_{\text{jeter}}, X_{\text{year}}] = \beta_0 + \beta_{\text{jeter}} X_{\text{jeter}} + \beta_{\text{year}} X_{\text{year}}$$

where the table gives approximately

$$\begin{aligned}
\beta_0 &= E[Y \mid X_{\text{jeter}} = 0, X_{\text{year}} = 0] = 104/411 = .253, \\
\beta_0 + \beta_{\text{jeter}} &= E[Y \mid X_{\text{jeter}} = 1, X_{\text{year}} = 0] = 12/48 = .250, \\
\beta_0 + \beta_{\text{year}} &= E[Y \mid X_{\text{jeter}} = 0, X_{\text{year}} = 1] = 45/140 = .321, \\
\beta_0 + \beta_{\text{jeter}} + \beta_{\text{year}} &\approx E[Y \mid X_{\text{jeter}} = 1, X_{\text{year}} = 1] = 183/582 = .314.
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The table also gives us

$$\begin{aligned}
E[X_{\text{year}} \mid X_{\text{jeter}} = 1] &= 582/630 = .924, \\
E[X_{\text{year}} \mid X_{\text{jeter}} = 0] &= 140/551 = .254.
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Both players hit better in 1996, and Jeter has more of his at bats that year. In the additive regression, this contributes to the coefficient when year is omitted:

$$\beta'_{\text{jeter}} = \beta_{\text{jeter}} + \beta_{\text{year}} \beta_{\text{jeter,year}}, \qquad \beta_{\text{jeter,year}} = .924 - .254 = .670.$$

Question

Recall the AR(1) process

$$Y_t = \rho Y_{t-1} + \epsilon_t.$$

Assume the process is stationary (all Y_t have the same marginal distribution). Suppose we regress Y_t on *two* previous values, Y_{t-1} and Y_{t-2}:

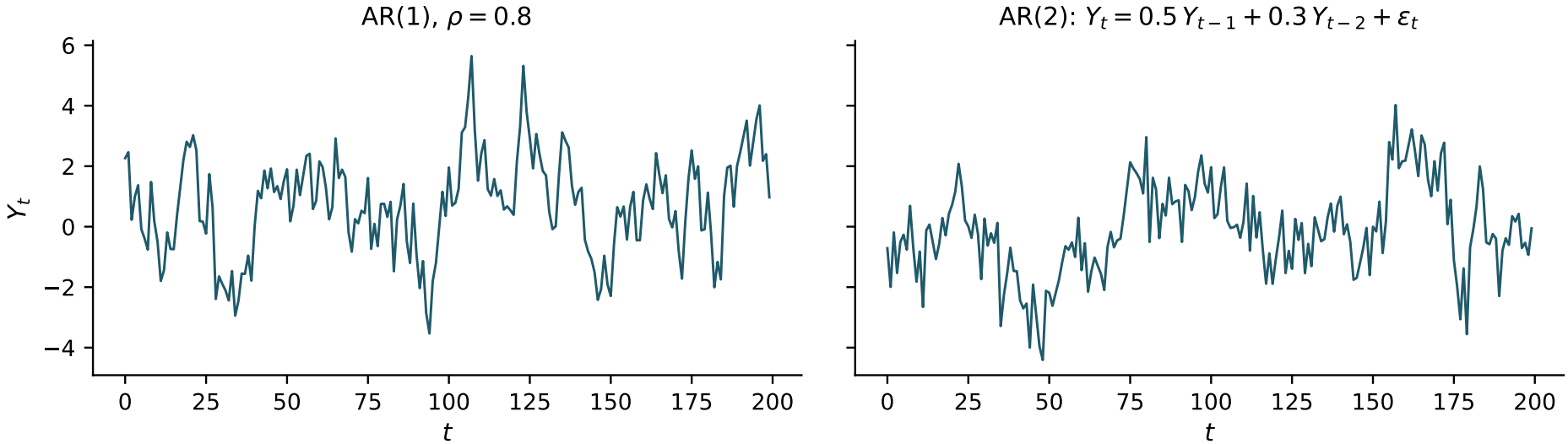
$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \epsilon_t'.$$

What should β_1 and β_2 be? Derive the regression coefficient when Y_{t-1} is omitted from the model using OVB. Then generalize this to obtain the regression coefficient between Y_{t-k} and Y_t.

An AR(2) process

Now let Y_t depend on *two* previous values:

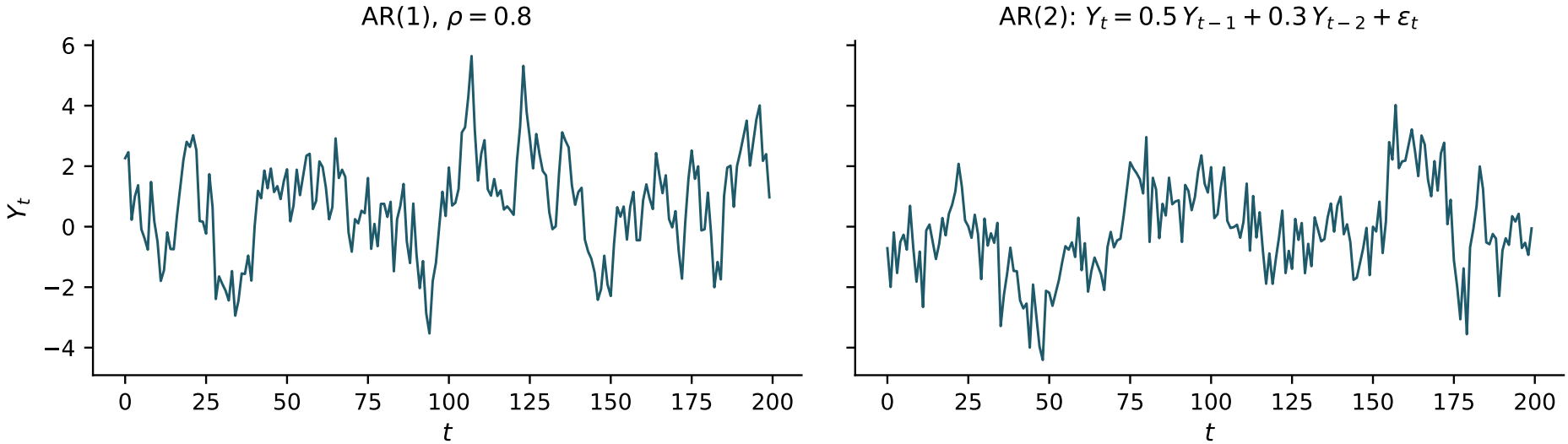
$$Y_t = 0.5\, Y_{t-1} + 0.3\, Y_{t-2} + \epsilon_t, \qquad \epsilon_t \sim \text{Normal}(0, 1).$$



An AR(2) process

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Question

What is the slope of the single-predictor regression of Y_t on Y_{t-1} (with Y_{t-2} excluded)?

Example

How big is the incumbency advantage? US House elections, 435 districts (from *Regression and Other Stories*):

- Y (`vote`): Democratic share of the two-party vote in 1988;
- X_{past} (`past_vote`): Democratic share in 1986;
- X_{inc} (`inc`): $+1$ if a Democrat is the incumbent, -1 if a Republican is, 0 for an open seat.

Uncontested races are set to 0.75 or 0.25.

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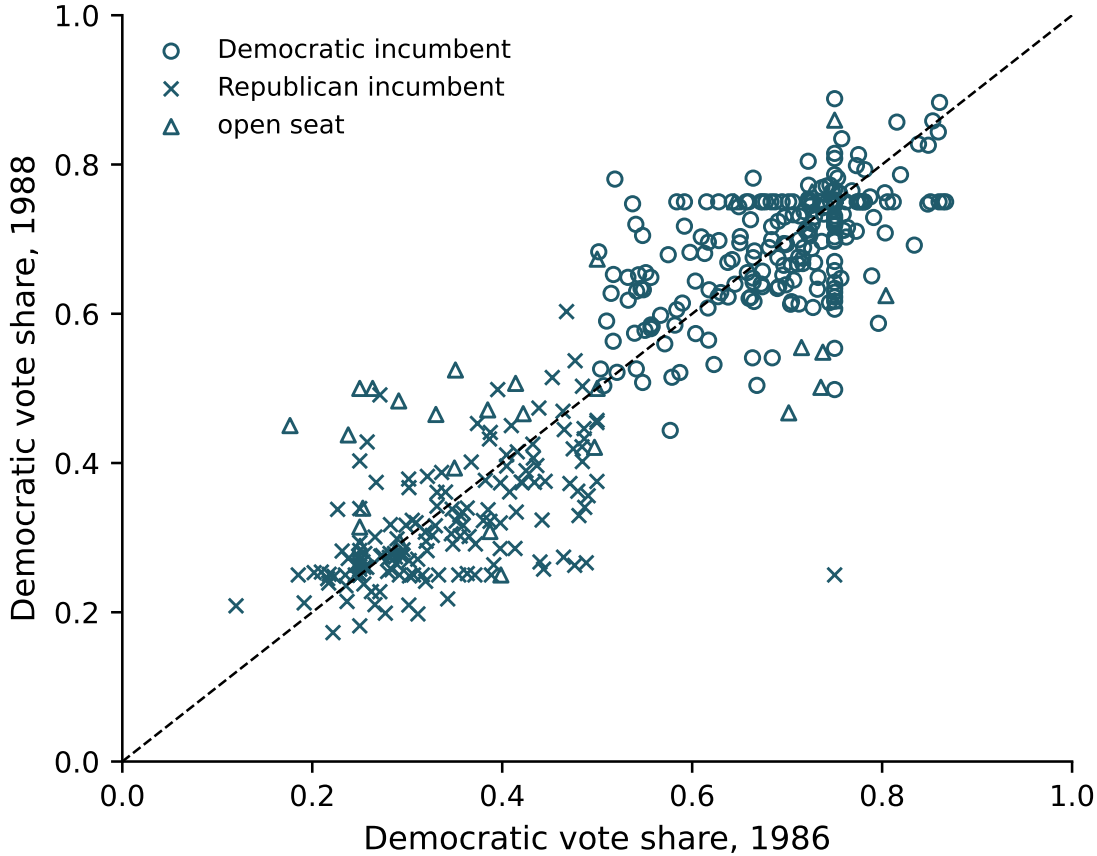
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Question

What is at least one limitation of this model?

1988 vs. 1986 vote



Each point is one district. The dashed line is $y = x$.

Dep. Variable:			vote	R-squared:		0.819
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.5063	0.004	121.695	0.000	0.498	0.514
inc	0.1906	0.004	44.210	0.000	0.182	0.199
Dep. Variable:			vote	R-squared:		0.887
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.2380	0.017	14.068	0.000	0.205	0.271
past_vote	0.5210	0.032	16.169	0.000	0.458	0.584
inc	0.0964	0.007	14.284	0.000	0.083	0.110

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Question

Interpret the coefficient on `inc` in each model. The slope of `past_vote` on `inc` is 0.181; check the omitted variable bias formula. Which factor is more important, incumbency or past vote?

Theory for $K \geq 2$ predictors

Theorem

Suppose $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_K X_K + \epsilon$, *where* ϵ *is uncorrelated with* $X_1, \ldots, X_K$.

Omitted variable bias: if $X_2, \ldots, X_K$ *are left out, the slope of the regression of* Y *on* X_1 *is*

$$\boxed{\beta_1' = \beta_1 + \sum_{j=2}^{K} \beta_j \beta_{1,j}}, \qquad \beta_{1,j} = \frac{cov(X_1, X_j)}{var(X_1)}.$$

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Question

Derive this formula, assuming $E[X_j] = 0$ for $j = 1, \ldots, K$ and $\beta_0 = 0$ (hence $E[Y] = 0$).

Test scores with three predictors

Children's test scores again, now adding X_{age} (mom_age), the mother's age at birth.

Dep. Variable:		kid_score	R-squared:			0.008
	coef	std err	t	P>|t|	[0.025	0.975]
const	70.9569	8.307	8.542	0.000	54.631	87.283
mom_age	0.6952	0.362	1.920	0.055	-0.016	1.407

Dep. Variable:		kid_score	R-squared:			0.215
	coef	std err	t	P>|t|	[0.025	0.975]
const	20.9847	9.130	2.298	0.022	3.039	38.930
mom_hs	5.6472	2.258	2.501	0.013	1.210	10.085
mom_iq	0.5625	0.061	9.276	0.000	0.443	0.682
mom_age	0.2248	0.331	0.680	0.497	-0.425	0.875

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Question

Interpret the coefficient on `mom_age` in each model. With only `mom_hs` and `mom_iq`, the coefficients were 5.95 and 0.564 and $R^2 = 0.214$. Why did adding `mom_age` change so little even though age is associated with test scores in the first regression? What does this tell us about association between age and the other predictors?

Mother's age vs. schooling and IQ

Dep. Variable:		mom_hs	R-squared:			0.046
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0423	0.164	0.258	0.797	-0.280	0.365
mom_age	0.0326	0.007	4.565	0.000	0.019	0.047
Dep. Variable:		mom_iq	R-squared:			0.008
	coef	std err	t	P>|t|	[0.025	0.975]
const	88.4081	6.105	14.482	0.000	76.409	100.407
mom_age	0.5087	0.266	1.912	0.057	-0.014	1.032

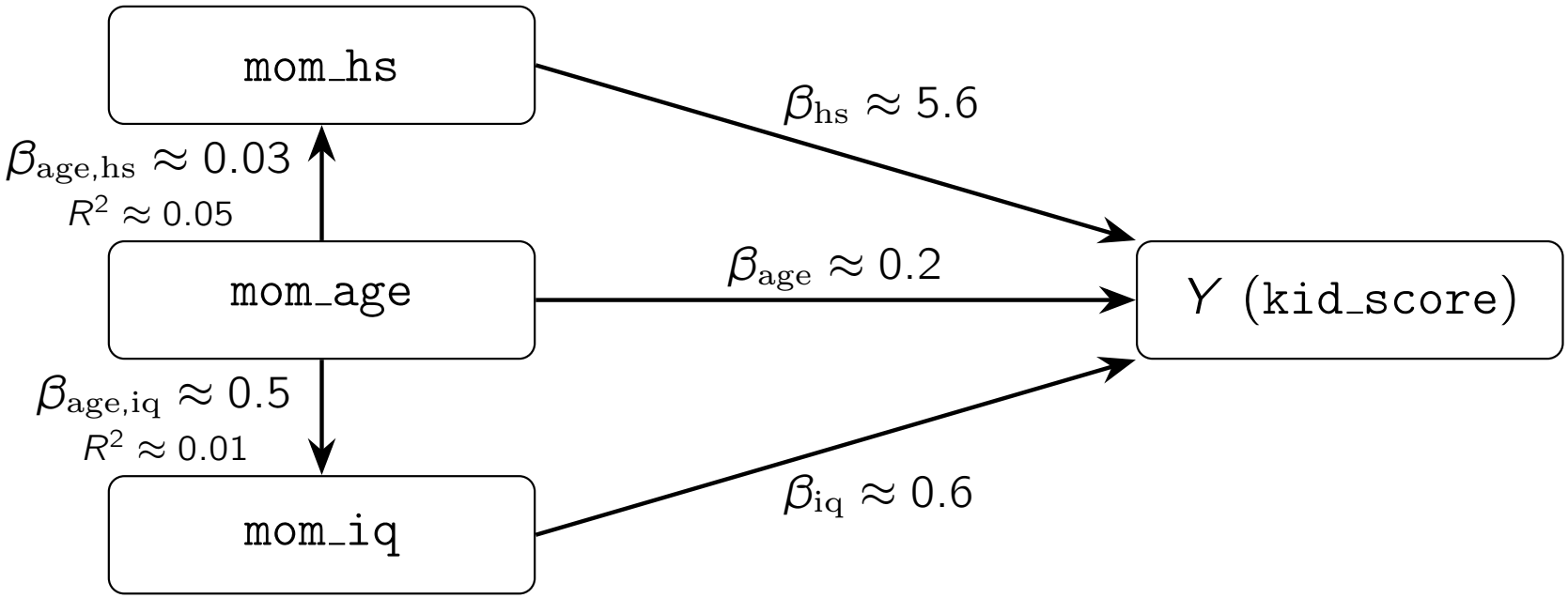
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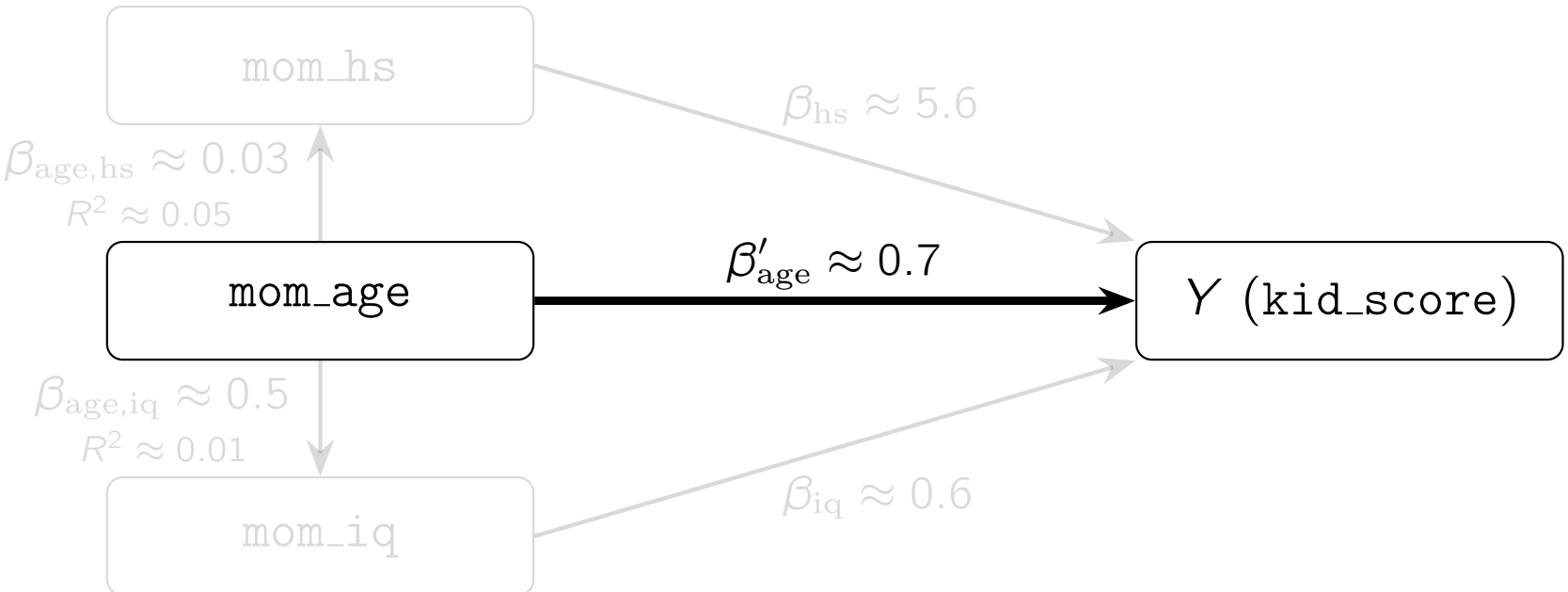
Question

Express the regression coefficient of the single predictor regression with age.

How mother's age is related to test scores



How mother's age is related to test scores



$$\underbrace{0.7}_{\beta'_{\text{age}}} \approx \underbrace{0.2}_{\beta_{\text{age}}} + \underbrace{0.03 \times 5.6}_{\beta_{\text{age,hs}}\beta_{\text{hs}} \,\approx\, 0.2} + \underbrace{0.5 \times 0.6}_{\beta_{\text{age,iq}}\beta_{\text{iq}} \,=\, 0.3}$$

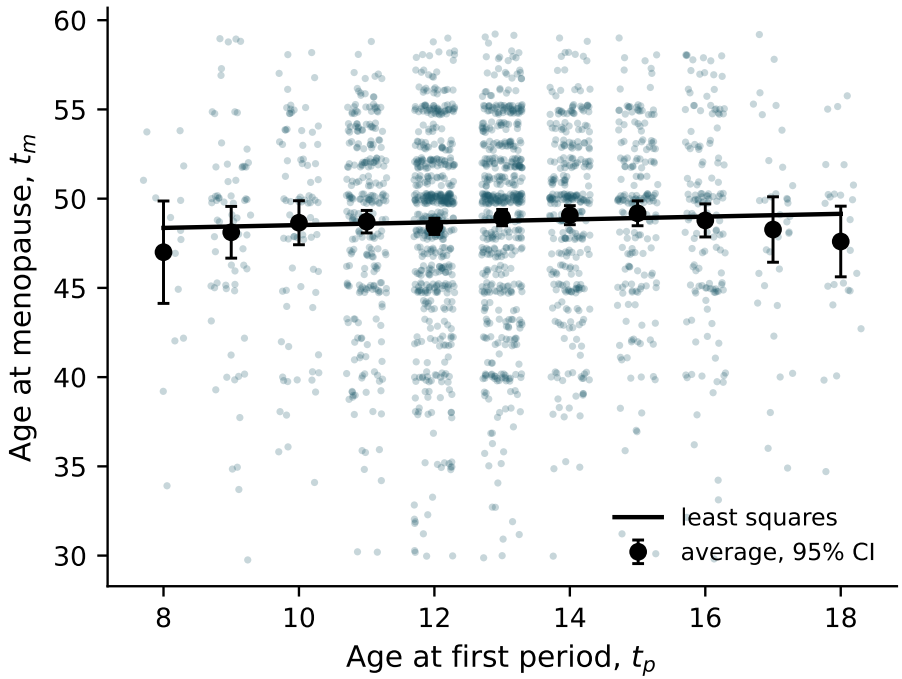
Example

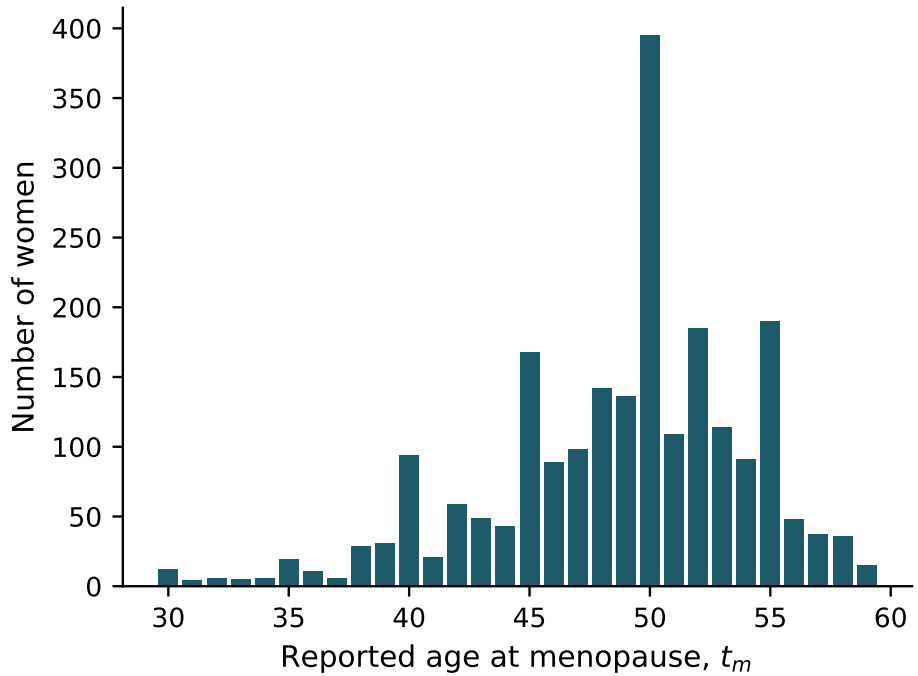
What factors influence the timing of menopause? Data from NHANES 2013–2018 survey of $N = 2248$ US women.

- t_m (`tm`): age at menopause (last period)
- t_p (`tp`): age at first period (puberty)
- X_{smoker} (`smoker`): 1 if she has smoked at least 100 cigarettes
- X_{births} (`births`): number of live births
- $X_{\text{birthyear}}$ (`birthyear`): year of birth

All ages are recalled at the interview.

t_m vs. t_p


Age at menopause, t_m
Age at first period, t_p
least squares
average, 95% CI


Number of women
Reported age at menopause, t_m

Dep. Variable:			tm	R-squared:		0.001
	coef	std err	t	P>|t|	[0.025	0.975]
const	47.7292	0.805	59.256	0.000	46.150	49.309
tp	0.0790	0.062	1.276	0.202	-0.042	0.200

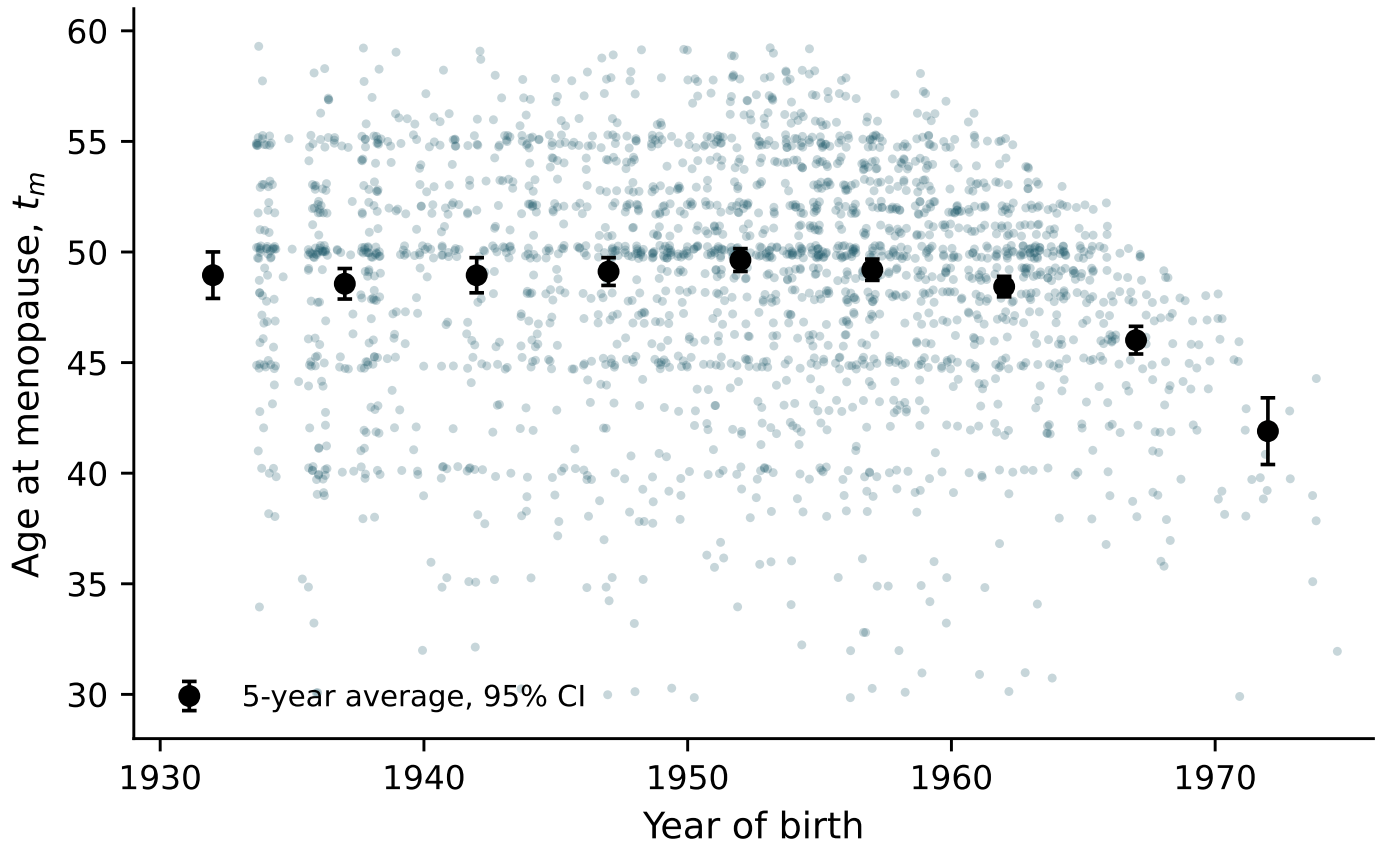
Dep. Variable:			tm	R-squared:		0.015
	coef	std err	t	P>|t|	[0.025	0.975]
const	165.0489	24.430	6.756	0.000	117.140	212.958
tp	0.0522	0.063	0.827	0.409	-0.072	0.176
smoker	-0.6673	0.239	-2.798	0.005	-1.135	-0.200
births	-0.1288	0.062	-2.066	0.039	-0.251	-0.007
birthyear	-0.0596	0.012	-4.778	0.000	-0.084	-0.035

```
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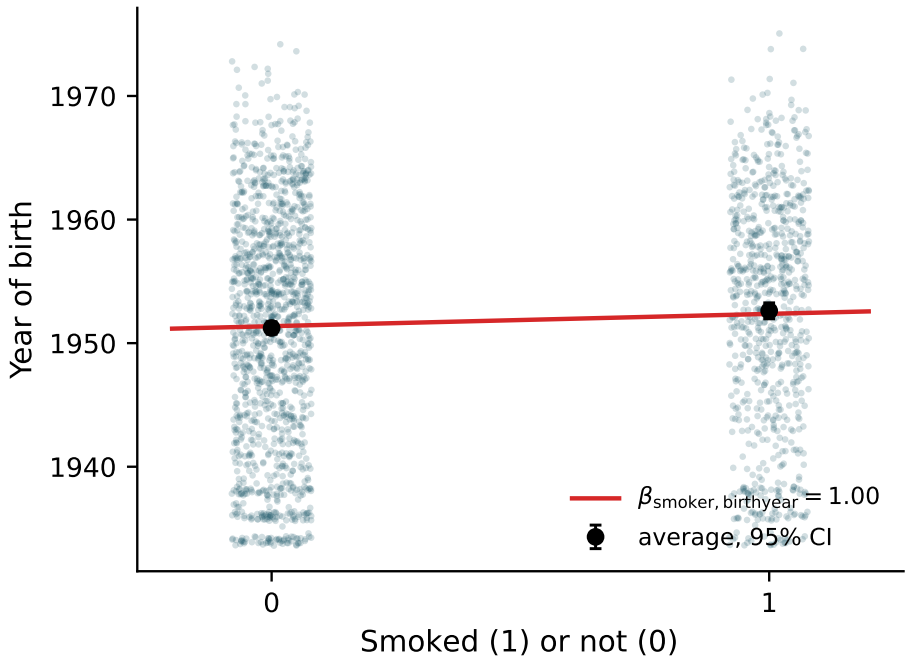
Question

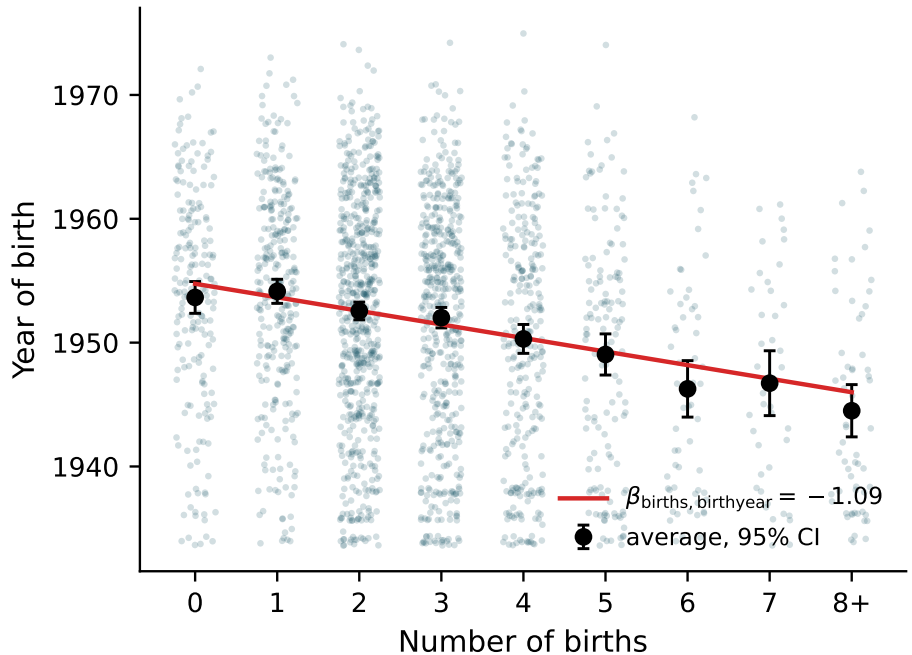
Interpret the coefficients. `smoker` and `birthyear` are clearly significant, yet $R^2 = 0.015$. What does that tell you?

t_m vs. year of birth



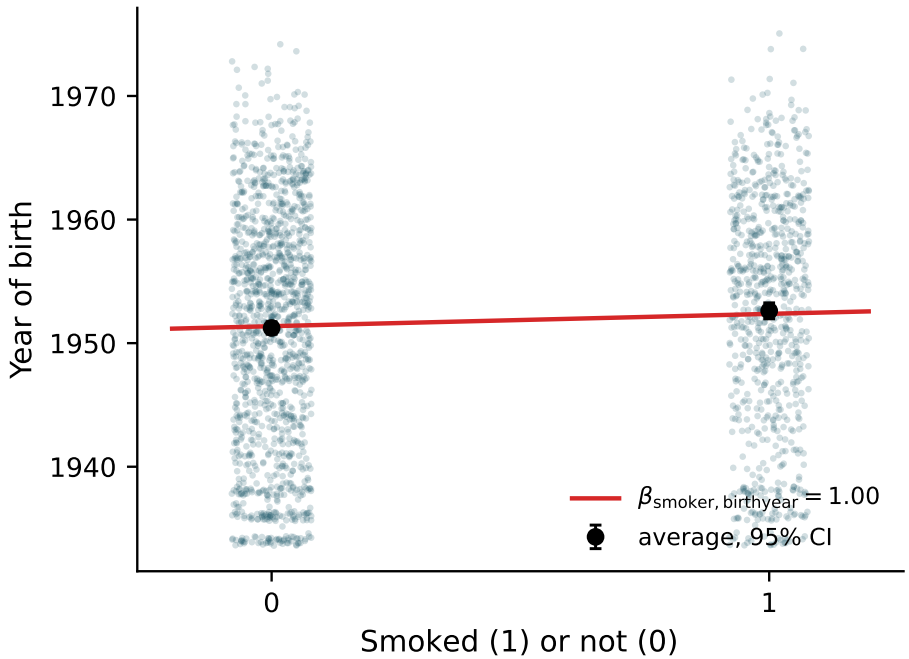
Year of birth vs. smoking and births

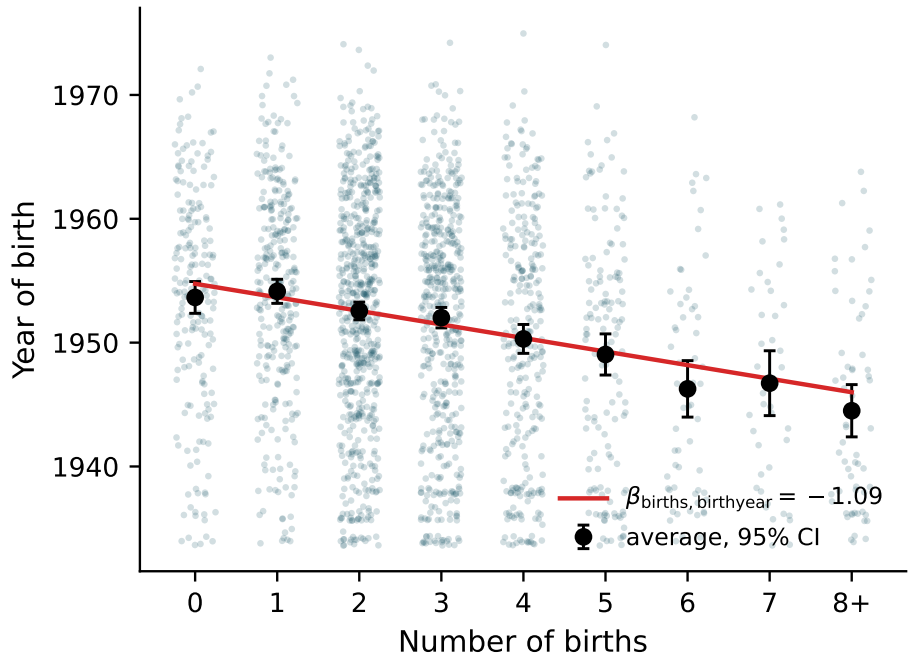




Black dots are averages with 95% confidence intervals. Red lines: regression of `birthyear` on `tp`, `smoker`, `births`, with the other predictors held at their means.

Year of birth vs. smoking and births



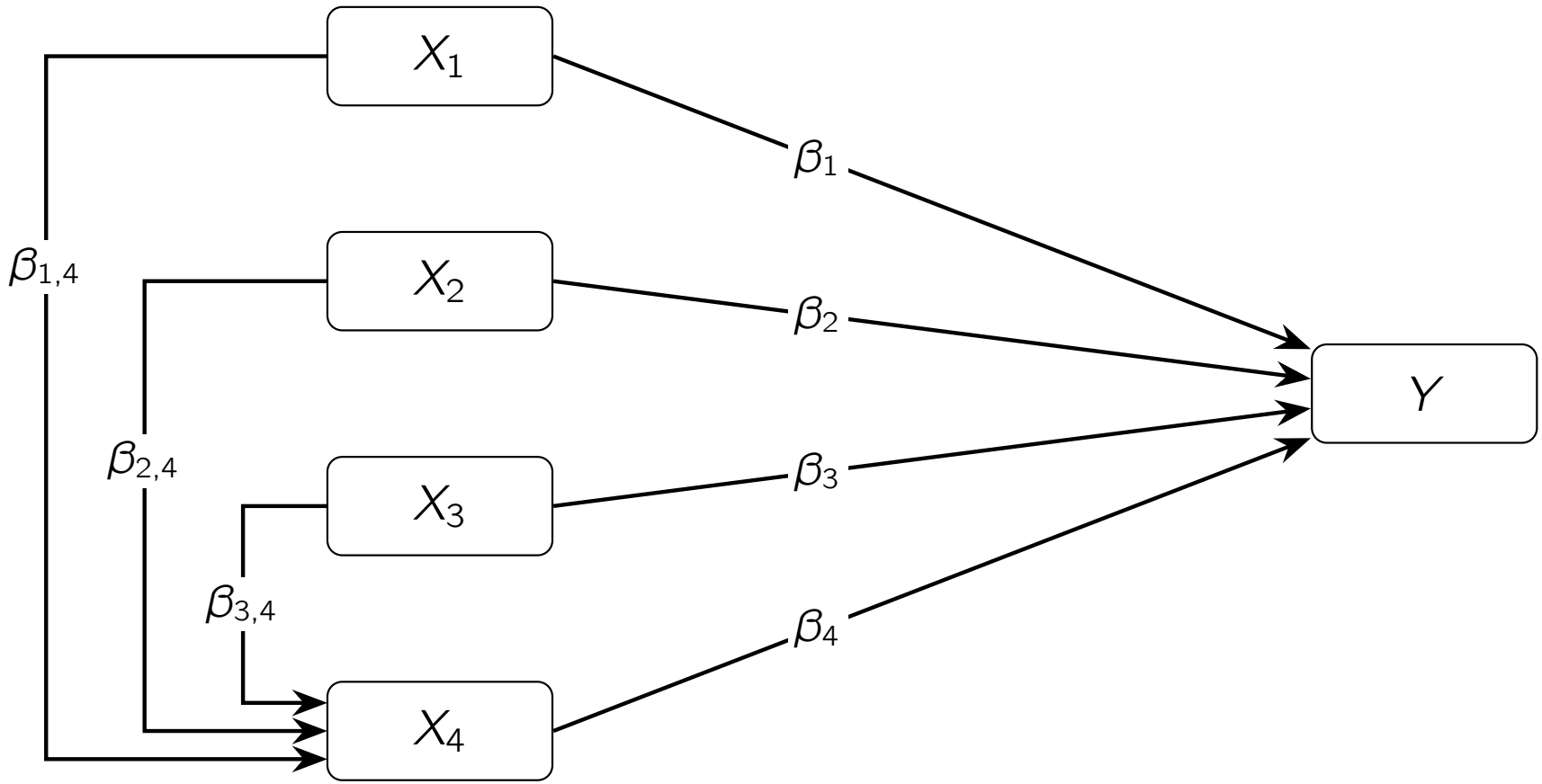


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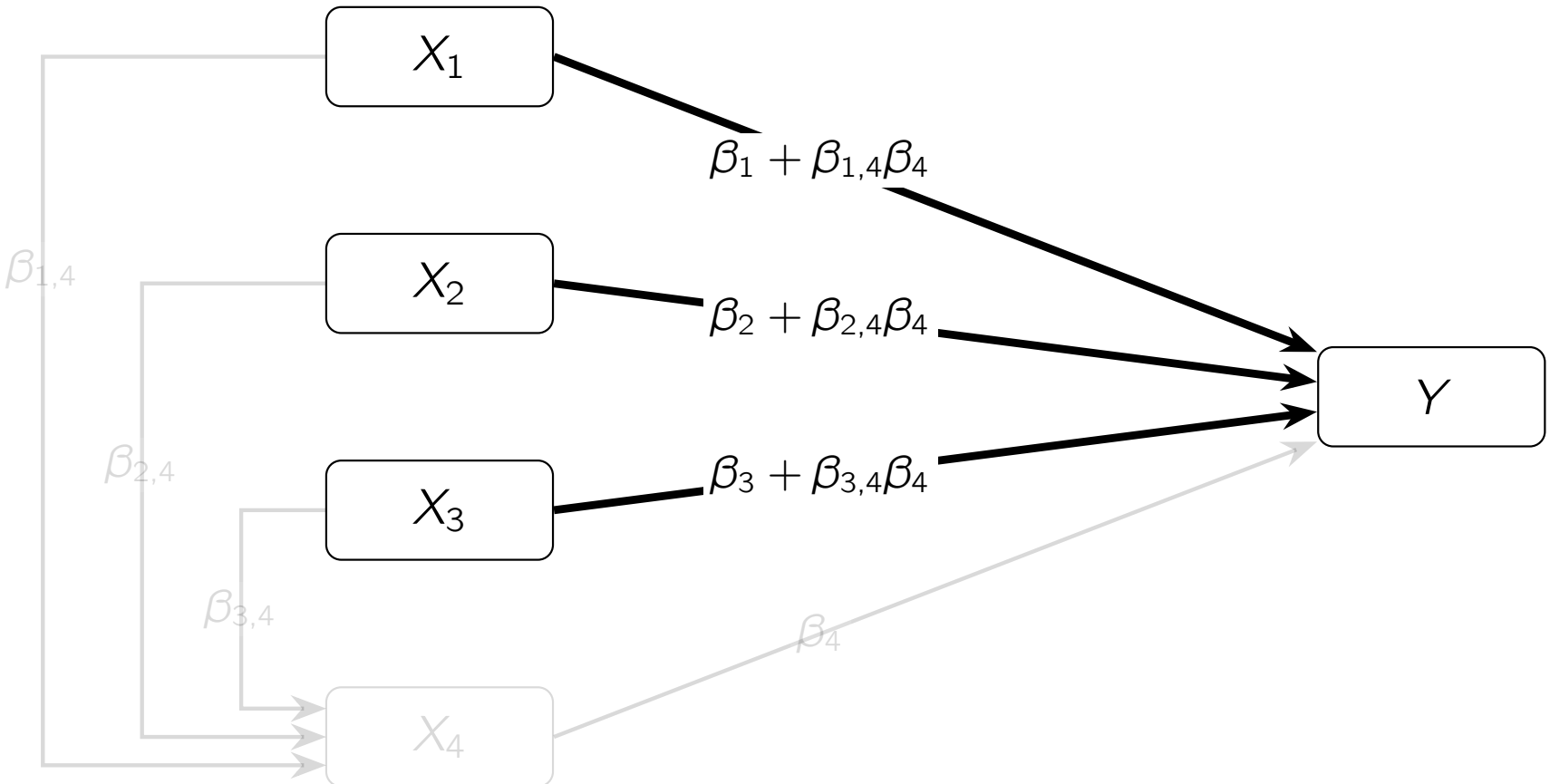
Question

Without refitting the model, can you predict how do these trends affect the coefficients on `smoker` and `births` when `birthyear` is left out of the model? (do they go up or down?)

Leaving out one predictor of many

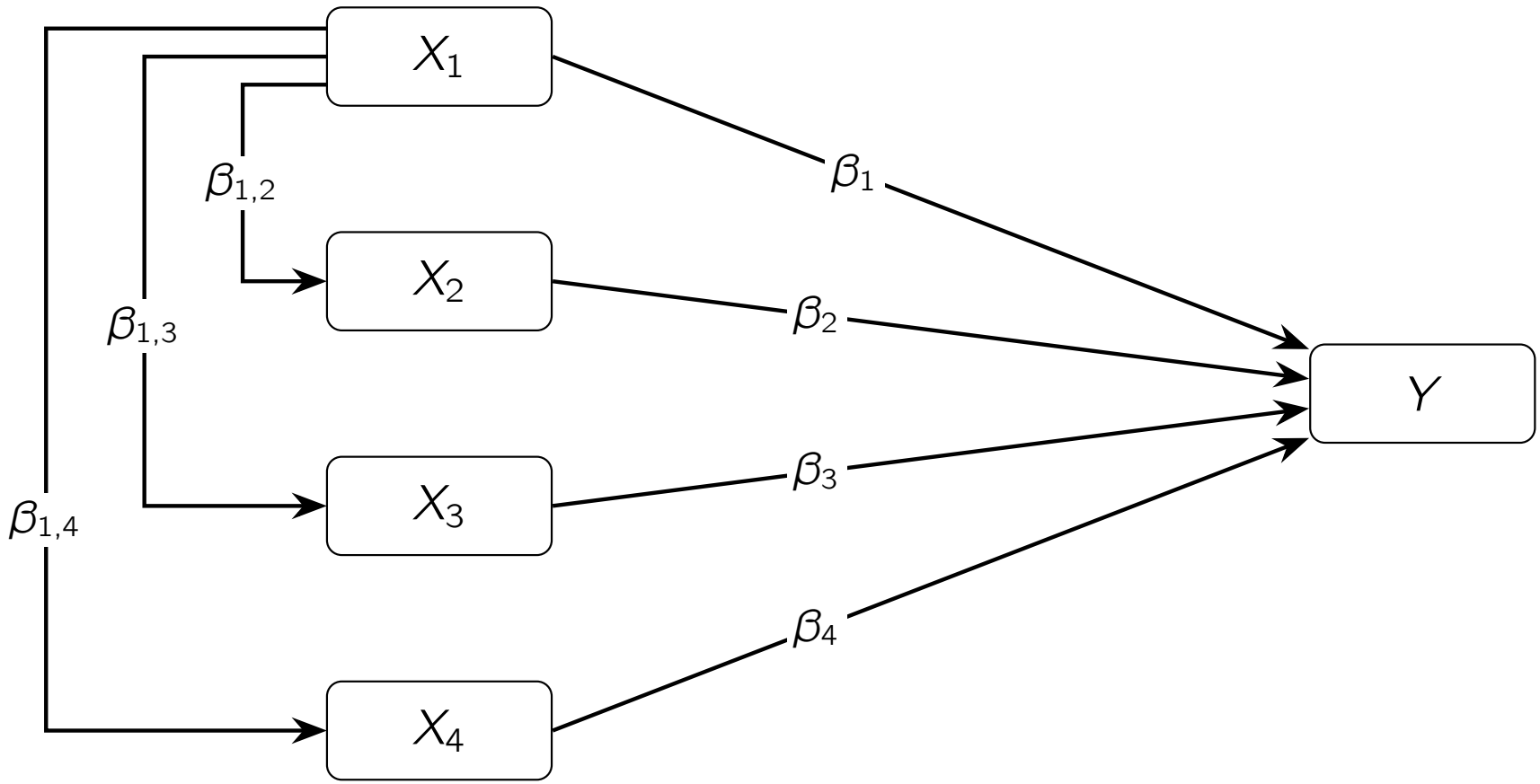


Leaving out one predictor of many

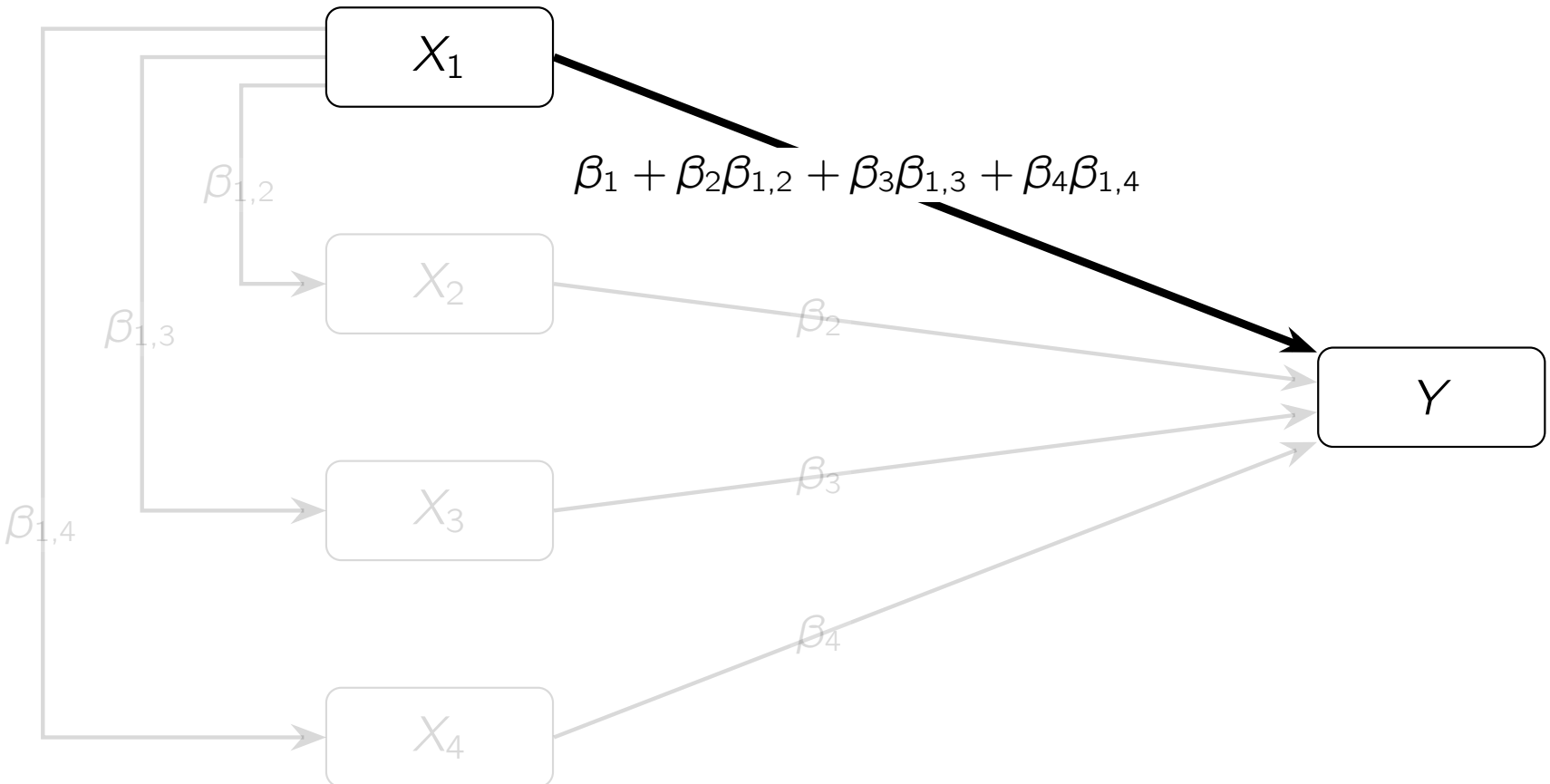


Each remaining coefficient picks up β_4 times the coefficient of its predictor in the regression of X_4 on X_1, X_2, X_3.

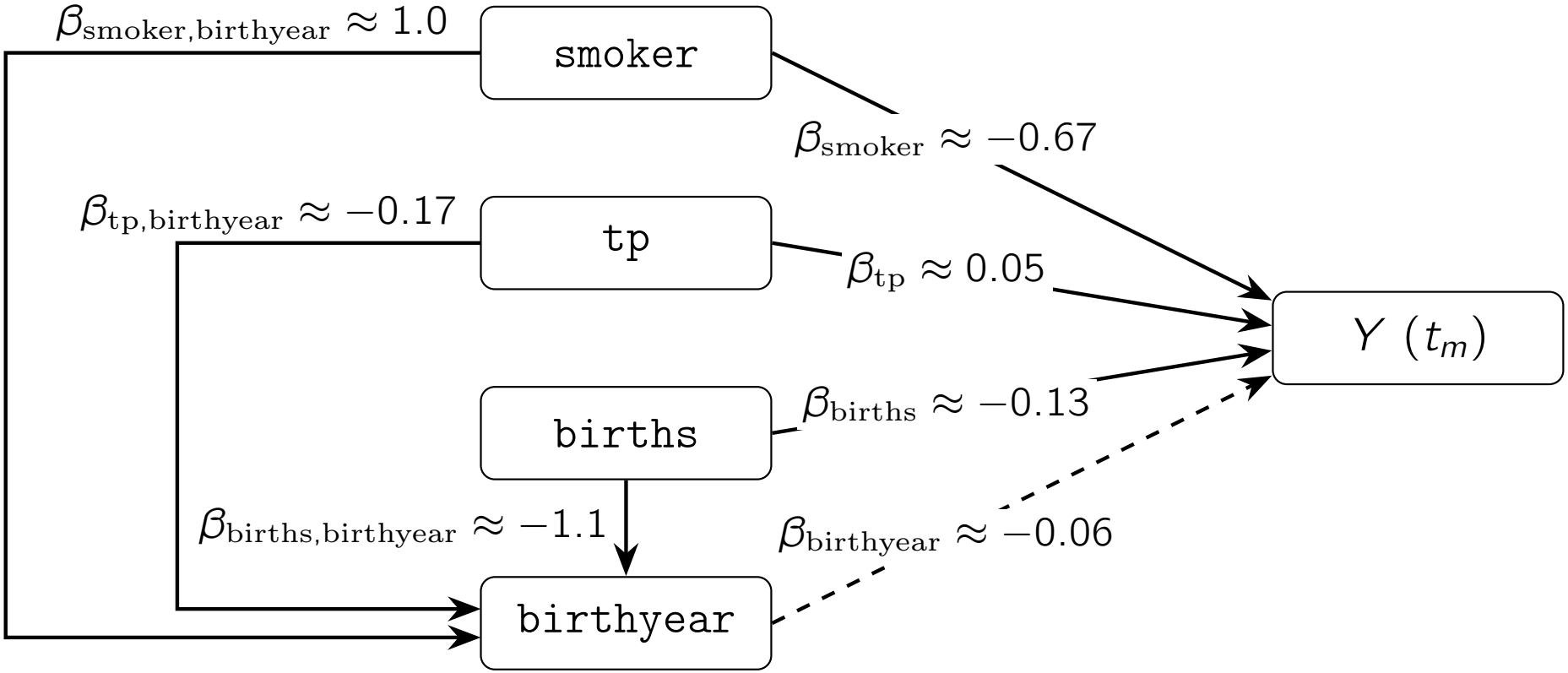
Compared to OVB Theorem from earlier



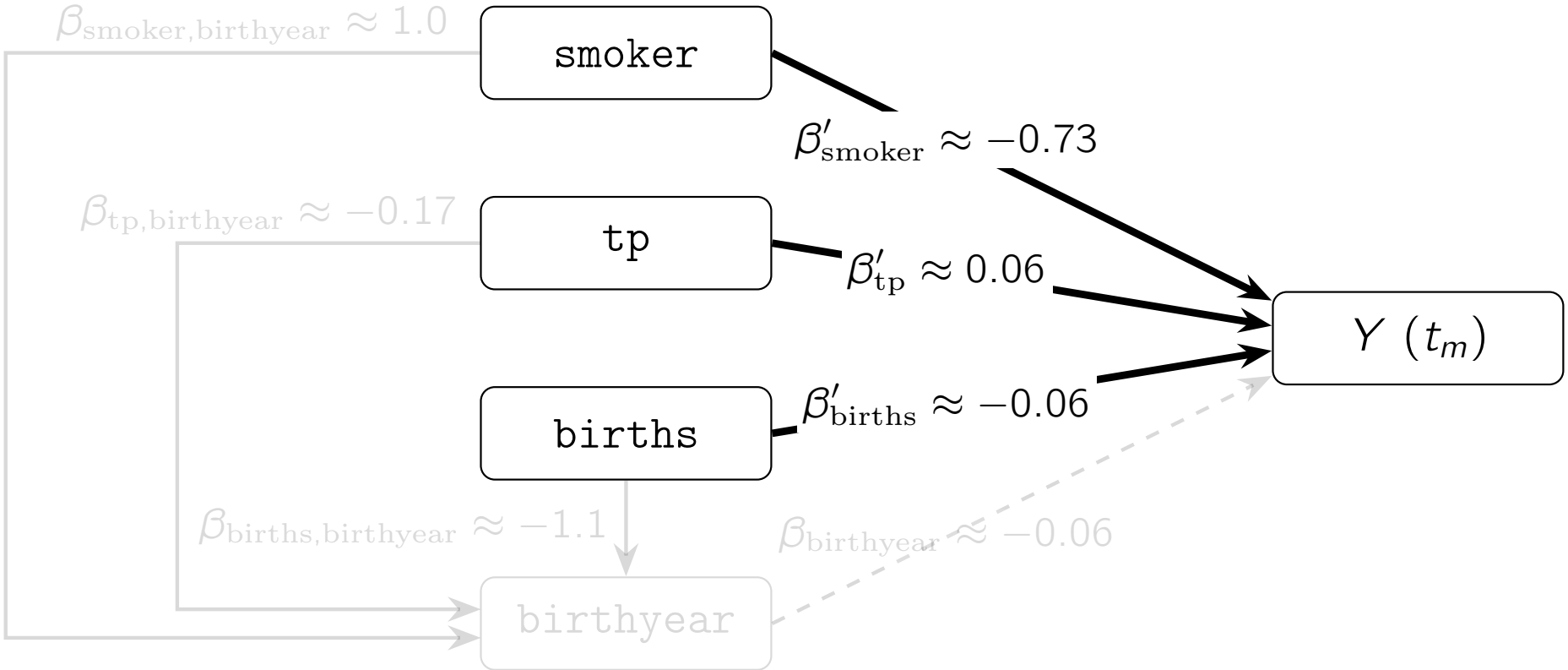
Compared to OVB Theorem from earlier



Here $\beta_{1,j}$ is the single-predictor slope of X_j on X_1: this is the theorem for K predictors.



How do the coefficients on `smoker` and `births` change if `birthyear` is left out?



How do the coefficients on `smoker` and `births` change if `birthyear` is left out?

$$\underbrace{-0.73}_{\beta'_{\text{smoker}}} \approx \underbrace{-0.67}_{\beta_{\text{smoker}}} + \underbrace{1.0 \times (-0.06)}_{\beta_{\text{smoker,birthyear}}\beta_{\text{birthyear}} \,=\, -0.06}$$

$$\underbrace{-0.06}_{\beta'_{\text{births}}} \approx \underbrace{-0.13}_{\beta_{\text{births}}} + \underbrace{(-1.1) \times (-0.06)}_{\beta_{\text{births,birthyear}}\beta_{\text{birthyear}} \,\approx\, 0.07}$$